

Simulating electronic systems with periodic boundary conditions

April 18, 2026

1 Mathematical identities

1.1 Special functions

The error function is defined

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x dt e^{-t^2}. \quad (1)$$

The complementary error function is

$$\operatorname{erfc}(x) = 1 - \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty dt e^{-t^2} = \frac{2}{\sqrt{\pi}} \int_0^{1/x} dt \frac{e^{-1/t^2}}{t^2}. \quad (2)$$

Useful limiting behaviors are

$$\operatorname{erf}(x) \rightarrow \begin{cases} 1 - \frac{e^{-x^2}}{\sqrt{\pi}x}, & (x \rightarrow \infty), \\ \frac{2x}{\sqrt{\pi}}, & (x \rightarrow 0). \end{cases} \quad (3)$$

1.2 Lattice things

Consider a 2D Bravais lattice defined by primitive vectors $\{\mathbf{a}_1, \mathbf{a}_2\}$. A general lattice vector is

$$\mathbf{R} = l_1 \mathbf{a}_1 + l_2 \mathbf{a}_2, \quad l_i \in \mathbb{Z}. \quad (4)$$

The primitive unit-cell area is $\Omega = |\mathbf{a}_1 \times \mathbf{a}_2|$. The reciprocal lattice is defined by $\{\mathbf{b}_1, \mathbf{b}_2\}$ such that $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$. A general reciprocal-lattice vector is

$$\mathbf{G} = l_1 \mathbf{b}_1 + l_2 \mathbf{b}_2, \quad l_i \in \mathbb{Z}. \quad (5)$$

A useful mathematical identity is

$$\sum_{\mathbf{G}} e^{i\mathbf{G} \cdot \mathbf{x}} = \Omega \sum_{\mathbf{R}} \delta(\mathbf{x} - \mathbf{R}). \quad (6)$$

This may be used to derive the “2D theta-transform” identity:

$$\sum_{\mathbf{R}} e^{-|\mathbf{r}-\mathbf{R}|^2 \xi^2} = \frac{\pi}{\Omega \xi^2} \sum_{\mathbf{G}} e^{-G^2/4\xi^2} e^{-i\mathbf{G}\cdot\mathbf{r}}. \quad (7)$$

Proof:

$$\sum_{\mathbf{R}} e^{-|\mathbf{r}-\mathbf{R}|^2 \xi^2} = \int d^2\mathbf{r}' e^{-r'^2 \xi^2} \sum_{\mathbf{R}} \delta(\mathbf{r}' - \mathbf{r} - \mathbf{R}) \quad (8)$$

$$= \frac{1}{\Omega} \sum_{\mathbf{G}} \int d^2\mathbf{r}' e^{-r'^2 \xi^2} e^{i\mathbf{G}\cdot(\mathbf{r}' - \mathbf{r})} \quad (9)$$

$$= \frac{1}{\Omega} \sum_{\mathbf{G}} e^{-i\mathbf{G}\cdot\mathbf{r}} \int d^2\mathbf{r}' e^{-r'^2 \xi^2} e^{i\mathbf{G}\cdot\mathbf{r}'} \quad (10)$$

$$= \frac{\pi}{\Omega \xi^2} \sum_{\mathbf{G}} e^{-i\mathbf{G}\cdot\mathbf{r}} e^{-G^2/4\xi^2}. \quad \square \quad (11)$$

2 Coulomb energy in a periodic system

Consider a system of N electrons with coordinates $\{\mathbf{r}_1, \dots, \mathbf{r}_N\}$ confined to a simulation cell. We impose periodic boundary conditions, which means our simulation cell is part of a larger “super-lattice.” We take our fundamental cell to be defined by primitive vectors $\{\mathbf{a}_1, \mathbf{a}_2\}$ and the super-lattice is spanned by $\mathbf{R} = l_1 \mathbf{a}_1 + l_2 \mathbf{a}_2$. The area of the primitive cell is $\Omega = |\mathbf{a}_1 \times \mathbf{a}_2|$.

The interaction energy of an electron at $\mathbf{r}_i$ with an electron at $\mathbf{r}_j$ (we assume $i \neq j$), together with its periodic images at $\mathbf{r}_j + \mathbf{R}$, is

$$E_{\text{pair}}(\mathbf{r}_i - \mathbf{r}_j) = \sum_{\mathbf{R}} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j - \mathbf{R}|} \quad (12)$$

$$= \frac{2\pi e^2}{\Omega} \sum_{\mathbf{G} \neq 0} \frac{e^{-i\mathbf{G}\cdot\mathbf{r}_{ij}}}{G}, \quad (\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j), \quad (13)$$

where $\mathbf{G}$ are reciprocal lattice vectors. We have used the identity (6). The term $\mathbf{G} = 0$ is removed by the interaction with the positive neutralizing background. The resulting sum converges slowly. To address this, write

$$\frac{1}{G} = \frac{\text{erf}(\epsilon G)}{G} + \frac{\text{erfc}(\epsilon G)}{G}. \quad (14)$$

Here ϵ is the adjustable “Ewald parameter.” The properties of erf and erfc are documented in 1.1. Importantly, erfc decays exponentially for large values of

its argument. We deal with the erf term in the following way:

$$\sum_{\mathbf{G} \neq 0} \frac{\text{erf}(\epsilon G)}{G} e^{-i\mathbf{G} \cdot \mathbf{r}_{ij}} = \frac{2}{\sqrt{\pi}} \int_0^\epsilon dt \sum_{\mathbf{G}} e^{-G^2 t^2} e^{-i\mathbf{G} \cdot \mathbf{r}_{ij}} - \frac{2\epsilon}{\sqrt{\pi}} \quad (15)$$

$$= \frac{2}{\sqrt{\pi}} \int_0^\epsilon dt \frac{\Omega}{4\pi t^2} \sum_{\mathbf{R}} e^{-|\mathbf{r}_{ij} - \mathbf{R}|^2 / 4t^2} - \frac{2\epsilon}{\sqrt{\pi}} \quad (16)$$

$$= \frac{\Omega}{2\pi} \sum_{\mathbf{R}} \frac{1}{|\mathbf{r}_{ij} - \mathbf{R}|} \frac{2}{\sqrt{\pi}} \int_0^{2\epsilon/|\mathbf{r} - \mathbf{R}|} dt \frac{e^{-1/t^2}}{t^2} - \frac{2\epsilon}{\sqrt{\pi}} \quad (17)$$

$$= \frac{\Omega}{2\pi} \sum_{\mathbf{R}} \frac{1}{|\mathbf{r}_{ij} - \mathbf{R}|} \text{erfc}\left(\frac{|\mathbf{r}_{ij} - \mathbf{R}|}{2\epsilon}\right) - \frac{2\epsilon}{\sqrt{\pi}}. \quad (18)$$

The first line uses the definition of erf, going from the first to second line uses the 2D theta-transform (7), and to obtain the last line we use the definition of erfc (2). Putting everything together gives

$$E_{\text{pair}}(\mathbf{r}_{ij}) = \frac{2\pi e^2}{\Omega} \sum_{\mathbf{G} \neq 0} \frac{\text{erfc}(\epsilon G)}{G} e^{-i\mathbf{G} \cdot \mathbf{r}_{ij}} + e^2 \sum_{\mathbf{R}} \frac{\text{erfc}(|\mathbf{r}_{ij} - \mathbf{R}|/2\epsilon)}{|\mathbf{r}_{ij} - \mathbf{R}|} - \frac{4\sqrt{\pi} e^2 \epsilon}{\Omega}. \quad (19)$$

To include the interaction energy of the an electron with its own periodic images, we consider the limit

$$E_{\text{self}} = \lim_{\mathbf{r} \rightarrow 0} \left[E_{\text{pair}}(\mathbf{r}) - \frac{e^2}{r} \right], \quad (20)$$

so that

$$E_{\text{self}} = \frac{2\pi e^2}{\Omega} \sum_{\mathbf{G} \neq 0} \frac{\text{erfc}(\epsilon G)}{G} + e^2 \sum_{\mathbf{R} \neq 0} \frac{\text{erfc}(R/2\epsilon)}{R} - \frac{4\sqrt{\pi} e^2 \epsilon}{\Omega} - \frac{e^2}{\sqrt{\pi} \epsilon}. \quad (21)$$

The total energy of the system (per unit cell), which is a function of the coordinate of all N particles, is then

$$E[\{\mathbf{r}_i\}] = \sum_{i < j}^N E_{\text{pair}}(\mathbf{r}_{ij}) + \frac{N}{2} E_{\text{self}}. \quad (22)$$

For the Ewald parameter we take

$$\epsilon = \frac{1}{2} \sqrt{\frac{\Omega}{\pi}}, \quad (23)$$

which gives a comparable rate of convergence for both the $\mathbf{R}$ and $\mathbf{G}$ sums.